

HSIU-CHI LIU

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Experience

Software Engineer, Trend Micro, Taipei

Oct. 2025 - Present

- Designed and implemented a reliable distributed system, improving the end-to-end microservice orchestration success rate to 99%
- Improved production observability across distributed microservices by instrumenting services with **OpenTelemetry** and standardizing tracing and metrics
- Built cloud-native backend infrastructure on **AWS** using **Terraform**, **Kubernetes**, **Helm**, **Prometheus**, and **Grafana** to support repeatable deployment and operational monitoring

Software Engineer Intern, Hewlett-Packard, Taipei

Jul. 2024 - Jul. 2025

- Built a RESTful data integration API using Go and Gin, enabling internal systems to exchange departmental data through a unified backend interface
- Improved internal knowledge retrieval by developing a Retrieval-Augmented Generation (RAG) database for faster access to department-specific information

Industry-Academia Research, NTU & MediaTek MARC, Taipei

Feb. 2024 - Jul. 2025

- Improved RISC-V Vector Extension workload performance by 2%–3% through the design and implementation of custom LLVM instructions
- Validated custom instruction behavior and performance impact with QEMU simulations for architecture-level analysis

Projects

Cometline (Personal Side Project)

- Built a desktop AI orchestration platform that abstracts multiple LLM providers behind a unified interface, enabling chat, memory, web search, streaming responses, and tool execution
- Enabled advanced AI workflows by **supporting coding agent capabilities** and **computer-use automation** for complex task execution across local desktop environments
- Unified **TUI**, **CLI**, and **desktop experiences** to provide consistent AI orchestration workflows across terminal and graphical environments

go-symphony (Personal Side Project)

- Reduced Go web service setup effort by building an LLM-friendly CLI with Cobra and Bubble Tea for opinionated Gin, PostgreSQL, Supabase, and sqlc scaffolding
- Accelerated full-stack project bootstrapping by adding out-of-the-box Supabase initialization and optional SvelteKit or Next.js frontend scaffolds

Personal Blog with Comment System (Personal Side Project)

- Built a full-stack blog comment system using Go, Gin, sqlc, and PostgreSQL to support persistent user discussions
- Enabled real-time user interaction by implementing WebSocket-based comment updates and OAuth-secured authentication

Open Source Contribution

KubeRay

Open-source Kubernetes operator designed to simplify the deployment and management of Ray applications.

- PR #4038**: Improved test maintainability by consolidating duplicated support utilities across KubeRay test files
- PR #4044**: Modernized Go test code by removing redundant range variable captures using Go 1.22 scoped iteration semantics

Temporal

Durable Execution platform that abstracts away the complexity of building distributed systems.

- PR #813**: Fixed Helm chart testing and added support for test pod annotations

Education

National Taiwan University, M.S. in Computer Science

Sep. 2023 - Jul. 2025

- Relevant Knowledge**: Advanced Compiler, Operating Systems Design and Implementation, Computer Vision with Deep Learning, Web Development

Skills

- Programming Languages**: **Golang** (primary); Python, C/C++, TypeScript (development experience)
- Technologies & Tools**: Docker, Kubernetes, Terraform, AWS, PostgreSQL, Svelte, React